

Introduction to Computer Science

Spring 2026
hgl235@nyu.edu

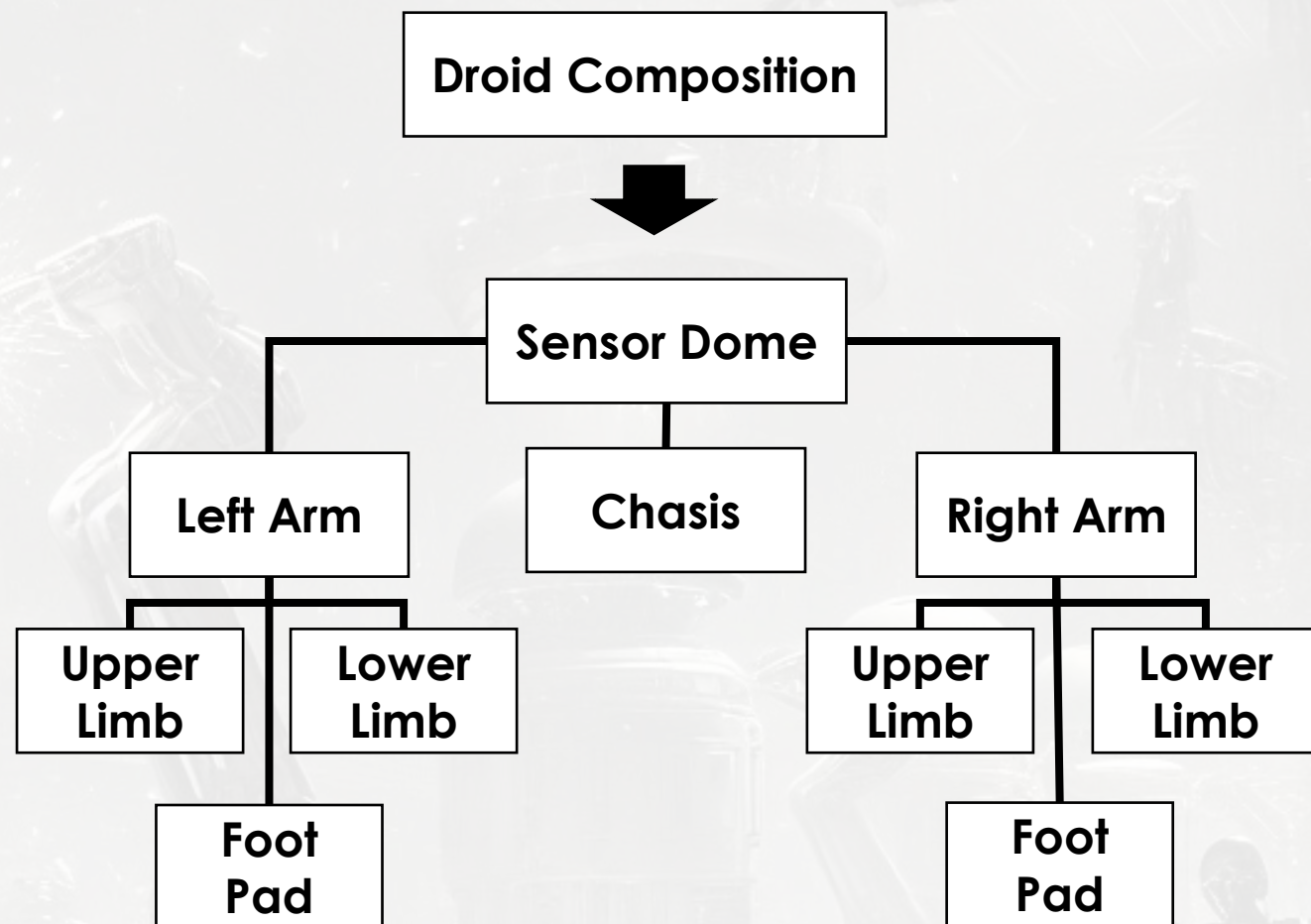
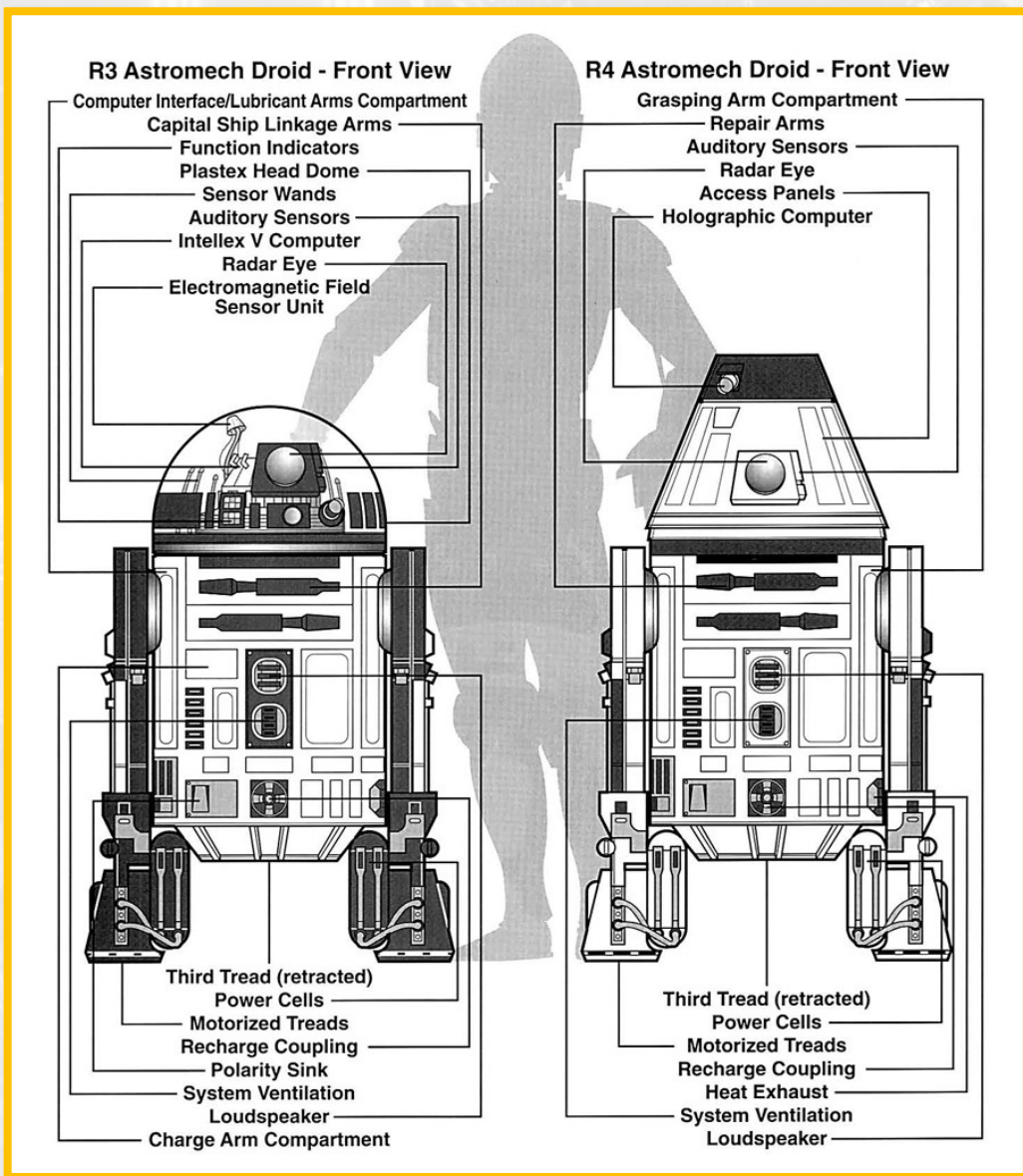


Homework 4

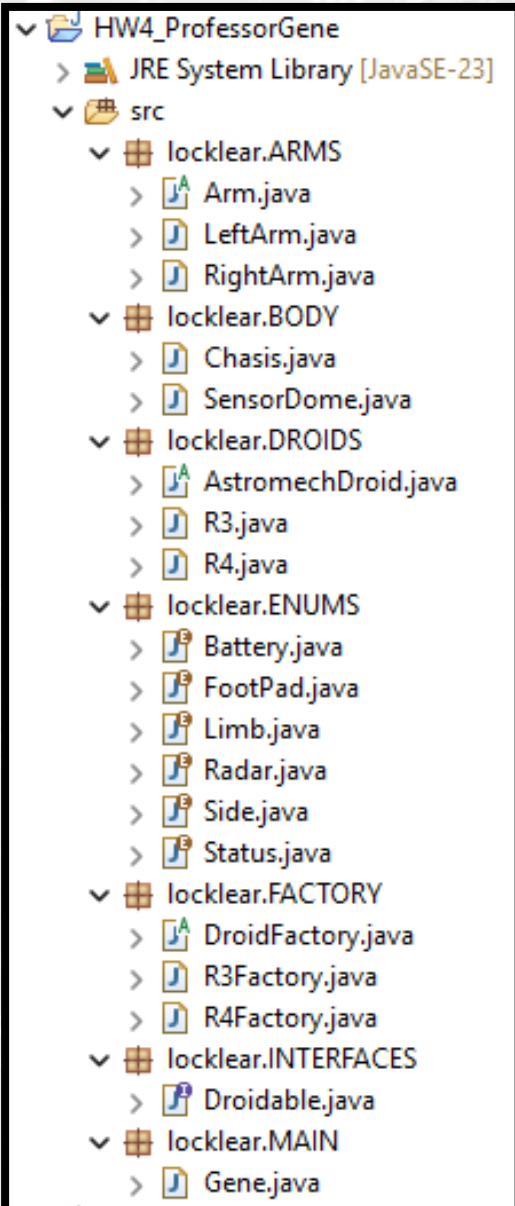
OOP Fundamentals

Professor HG Locklear

General



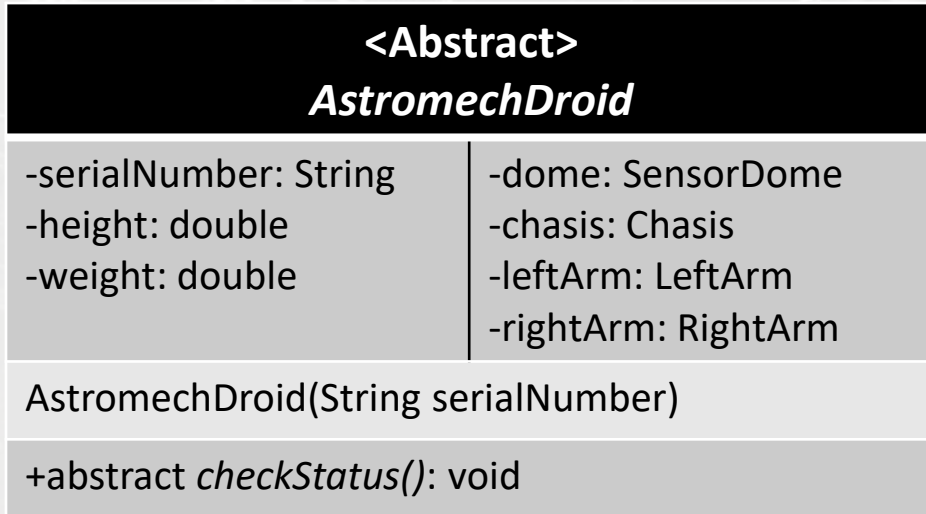
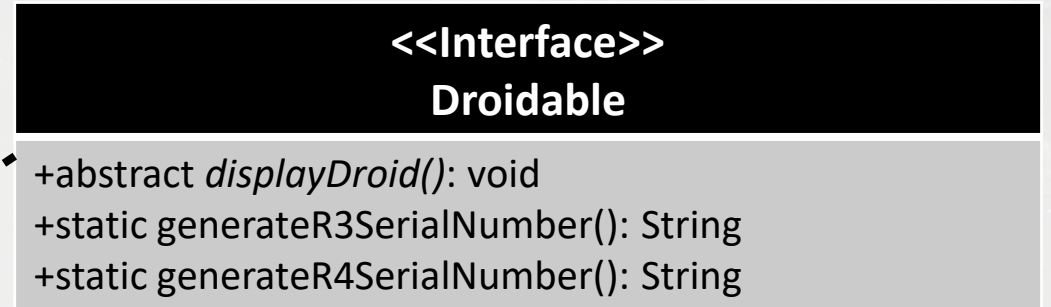
Program Structure



Create your Program Structure and main method EXACTLY as shown.

```
public class Gene {  
  
    public static void main(String[] args) throws IOException {  
  
        R3Factory R3 = new R3Factory("GR3");  
        R3.buildDroids(2);  
        R3.displayDroids();  
        System.out.println("*****");  
        System.out.println();  
        R4Factory R4 = new R4Factory("GR4");  
        R4.buildDroids(2);  
        R4.displayDroids();  
  
    }  
  
}
```

AstromechDroid Class



AT INITIALIZATION:

R3 Height is SensorDome Height + Chasis Height
R4 Height is SensorDome Height + Chasis Height

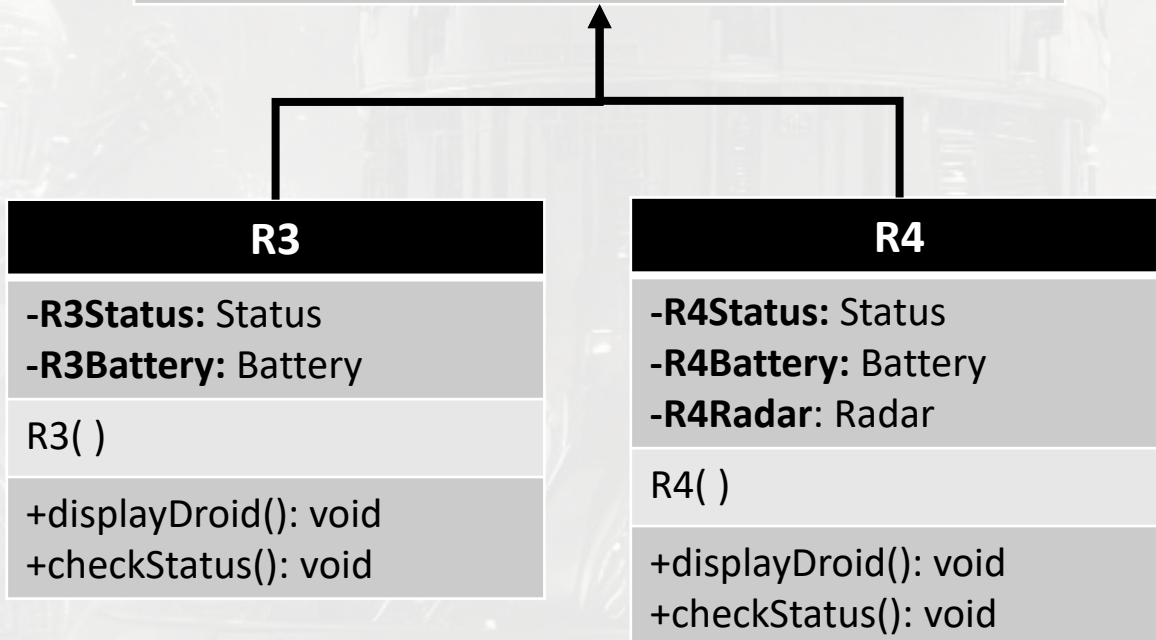
R3 Weight is SensorDome Weight + Chasis Weight
R4 Weight is SensorDome Weight + Chasis Weight

R3Status and R4Status are 'ONLINE'

R3Battery is R3
R4Battery is R4

R4Radar is R4R

ALL subclasses **must use** their superclass constructor



Chasis Class

Chasis
-serialNumber: String
-height: double
-weight: double
-status: Status
Chasis(String serialNumber)
+chasisCheck(): boolean

AT INITIALIZATION:
Status is **ONLINE**

serialNumber = “CH-” + serial Number of the AstromechDroid

AstromechDroid	Height	Weight
R3	4	400
R4	5	500

SensorDome Class

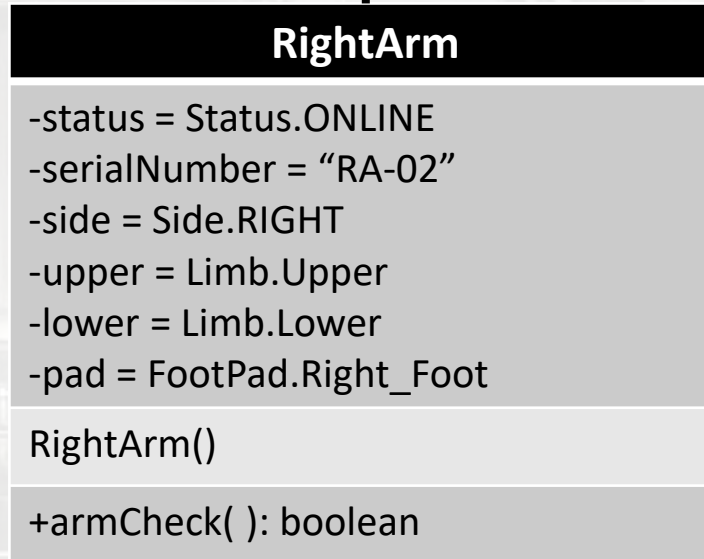
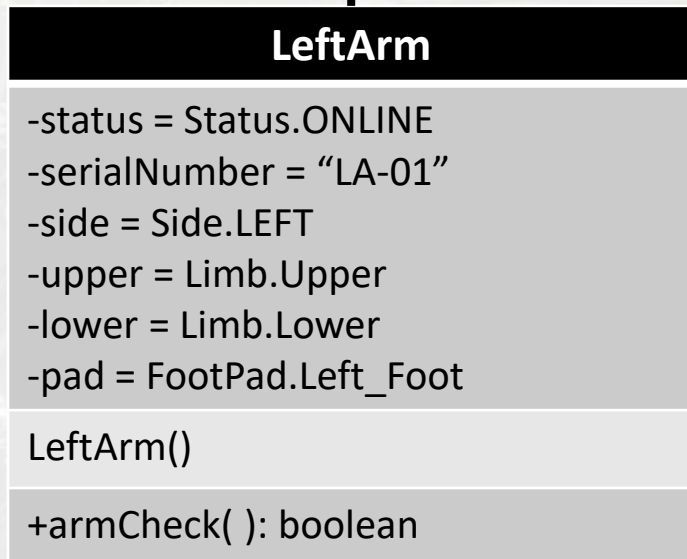
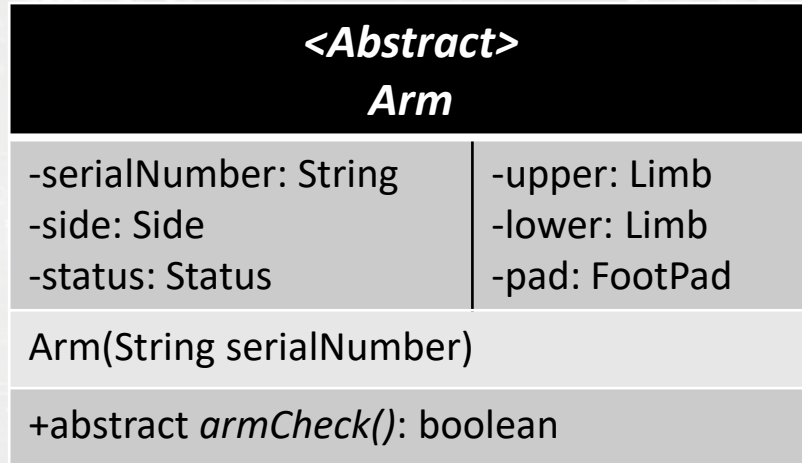
SensorDome
-serialNumber: String
-height: double
-weight: double
-status: Status
SensorDome(String serialNumber)
+sensorDomeCheck(): boolean

AT INITIALIZATION:
Status is **ONLINE**

serialNumber = “SD-” + serial Number of the AstromechDroid

AstromechDroid	Height	Weight
R3	2	200
R4	2	150

Arm Class



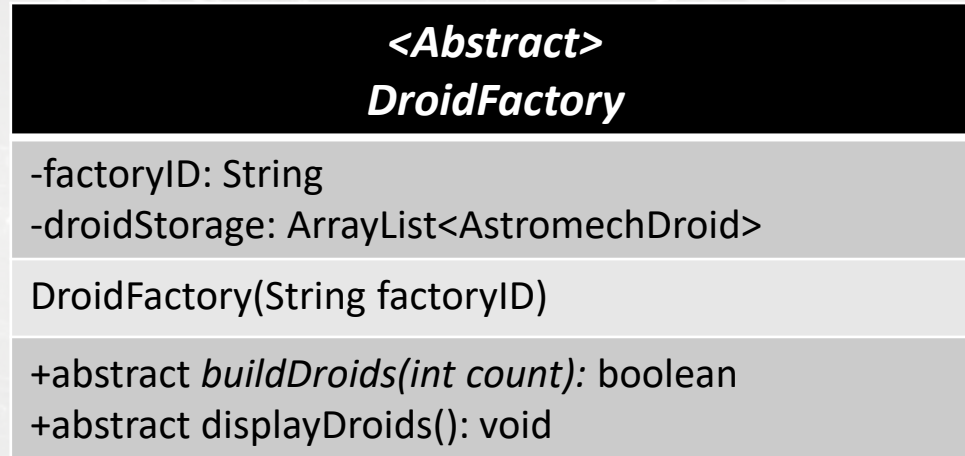
ALL subclasses **must use** their superclass constructor.

The initialization of **ALL** Object State **must occur** inside the subclass constructor.

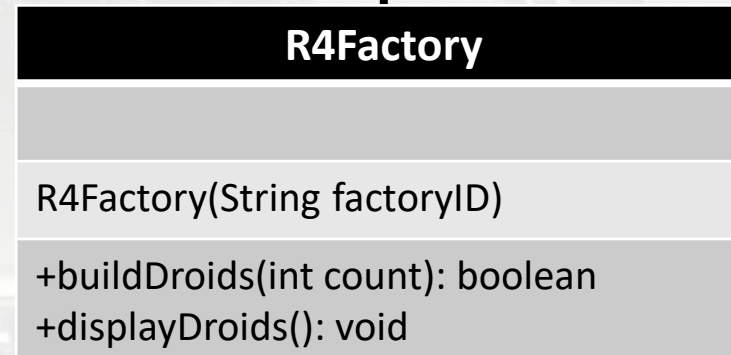
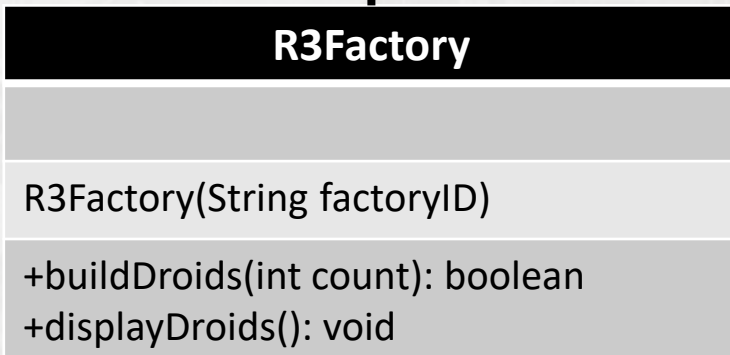
Enumerations

<ENUM> Side	<ENUM> Status	<ENUM> FootPad
LEFT,RIGHT	ONLINE,OFFLINE	LEFT_FOOT,RIGHT_FOOT
<ENUM> Limb	<ENUM> Battery	<ENUM> Radar
Upper("Upper",1) Lower("Lower",2)	R3("Lithium",1000) R4("Trithium",2000)	R4R("Doppler",500)
-type: String -value: int	-type: String -amps: int	-type: String -range: int
-Limb(String type, int value)	-Battery(String type, int amps)	-Radar(String type, int range)

DroidFactory Class



ALL subclasses **must use** their superclass constructor.



Method Specifications

Class or Interface	Method	Input	Processing
R3	displayDroid	None	Displays all information about the R3 droid as shown in the Expected Console Output
R4	displayDroid	None	Displays all information about the R4 droid as shown in the Expected Console Output
R3	checkStatus	None	Checks that the R3Status of R3 droid and prints 'R3 Droid is ONLINE' or 'R3 Droid is OFFLINE' to the console
R4	checkStatus	None	Checks that the R4Status of R4 droid and prints 'R4 Droid is ONLINE' or 'R4 Droid is OFFLINE' to the console
LeftArm	armCheck	None	Checks that status of the LeftArm Status is ONLINE if so, returns true false otherwise
RightArm	armCheck	None	Checks that status of the RightArm Status is ONLINE if so, returns true false otherwise
Chasis	chasisCheck	None	Checks that the status of the Chasis is ONLINE if so, returns true false otherwise
SensorDome	sensorDomeCheck	None	Checks that the status of the SensorDome is ONLINE if so, returns true false otherwise
Droidable	static generateR3SerialNumber	None	Returns a random string of the format "R3-####" (# is a single digit between 1 and 9)
Droidable	static generateR4SerialNumber	None	Returns a random string of the format "R4-####" (# is a single digit between 1 and 9)

Method Specifications

Class	Method	Input	Processing
R3Factory	buildDroids	count	Creates the specified number of R3 droids and adds them to the droidStorage attribute and returns true
R4Factory	buildDroids	count	Creates the specified number of R4 droids and adds them to the droidStorage attribute and returns true
R3Factory	displayDroids	None	Call the displayDroid method for all R3 droids in the droidStorage attribute
R4Factory	displayDroids	None	Call the displayDroid method for all R4 droids in the droidStorage attribute

Console Output

Problems @ Javadoc Declaration Console X
<terminated> Gene (21) [Java Application] C:\Users\msgth\p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_23.0.1.v2024102

```
R3 Droid          Height: 6.0      Weight: 600.0
Serial Number: R3-648
Dome Serial Number:      SD-R3-648  ONLINE
Chasis Serial Number:    CH-R3-648  ONLINE
Left Arm Serial Number:  LA-01    ONLINE
Right Arm Serial Number: RA-02    ONLINE
Battery: R3
Status: ONLINE
```

```
R3 Droid          Height: 6.0      Weight: 600.0
Serial Number: R3-524
Dome Serial Number:      SD-R3-524  ONLINE
Chasis Serial Number:    CH-R3-524  ONLINE
Left Arm Serial Number:  LA-01    ONLINE
Right Arm Serial Number: RA-02    ONLINE
Battery: R3
Status: ONLINE
```

```
R4 Droid          Height: 7.0      Weight: 650.0
Serial Number: R4-6628
Dome Serial Number:      SD-R4-6628  ONLINE
Chasis Serial Number:    CH-R4-6628  ONLINE
Left Arm Serial Number:  LA-01    ONLINE
Right Arm Serial Number: RA-02    ONLINE
Battery: R4
Status: ONLINE
Radar: R4R
```

```
R4 Droid          Height: 7.0      Weight: 650.0
Serial Number: R4-1415
Dome Serial Number:      SD-R4-1415  ONLINE
Chasis Serial Number:    CH-R4-1415  ONLINE
Left Arm Serial Number:  LA-01    ONLINE
Right Arm Serial Number: RA-02    ONLINE
Battery: R4
Status: ONLINE
Radar: R4R
```